

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 5

Case Study: Structural Breaks in Brazilian Inflation: Detecting Regime Changes with Bai–Perron and IIS

Application area: *Finance / Macroeconomics — Inflation Dynamics*

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IBGE / FRED: BRACPIALLMINMEI (Brazil CPI, monthly, 1980–2024). The 1990s hyperinflation and the 1994 Real Plan provide dramatic structural breaks ideal for illustrating the techniques of Lecture 5.

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1. Background and Objectives

Brazilian inflation history — from chronic high inflation through hyperinflation (1988–1994) to the Real Plan stabilisation — is one of the richest natural experiments in structural break econometrics. This tutorial applies the tools of Lecture 5 to this canonical dataset.

Learning objectives: (i) Detect structural breaks with Chow, CUSUM, and Bai–Perron tests; (ii) apply IIS for outlier-robust estimation; (iii) estimate a TVP state-space model for evolving inflation persistence; (iv) link structural change to economic history.

2. Data Preparation

```
library(strucchange); library(KFAS); library(fredr)
infl <- fredr("BRACPIALLMINMEI", frequency="m")
dinfl <- diff(log(infl$value)) * 100 # monthly inflation rate
```

3. Exercises

3.1. Visual analysis

Plot monthly Brazilian inflation 1980–2024. Identify visually: (a) the hyperinflation episode; (b) the Real Plan stabilisation (July 1994); (c) the 2002 BRL crisis. Mark these events on the plot. Does inflation look like an $I(1)$ series before 1994? After?

3.2. Bai–Perron multiple break test

Apply the Bai–Perron test (**breakpoints** in R's **strucchange**) to a regression of inflation on a constant over 1980–2024.

- How many breaks are selected? Report the estimated break dates and 95% confidence intervals.
- Compute the mean inflation within each regime.
- Match each regime to Brazil's economic history (Sarney, Collor, Itamar/Real Plan, Lula/Dilma, etc.).

3.3. CUSUM and CUSUMSQ

Estimate an AR(2) for monthly inflation 1980–1993 (pre-stabilisation). Plot CUSUM and CUSUMSQ of recursive residuals. Does the CUSUM cross its 5% boundary? At approximately what date? What does CUSUMSQ reveal about variance stability?

3.4. Impulse Indicator Saturation

Using the `gets` package in R, apply IIS to an AR(2) for inflation over the full 1980–2024 sample with target size $\alpha = 0.01$.

- (a) How many impulse dummies are retained? Report the dates and magnitudes.
- (b) Do the retained dummies correspond to known economic events?
- (c) Compare the estimated AR coefficients with and without IIS. How does outlier cleaning affect persistence estimates?

3.5. Time-varying persistence (Kalman filter)

Estimate a TVP AR(1) for inflation using the Kalman filter (KFAS package):

$$\pi_t = \beta_{0,t} + \beta_{1,t}\pi_{t-1} + \varepsilon_t, \quad \beta_t = \beta_{t-1} + \eta_t.$$

Plot the filtered estimate of $\beta_{1,t}$ (inflation persistence) over 1980–2024. Describe how persistence changed across the hyperinflation, stabilisation, and inflation targeting periods.

4. Core References

Bai, J. & Perron, P. (1998). *Econometrica*, 66, 47–78. Harvey, A.C. (1993). *Time Series Models*. MIT Press. Johansen, S. & Nielsen, B. (2016). *Scand. J. Stat.*, 43, 321–348.